Lavish Avinash

Software Engineer | Backend Systems Architect | AI & Agentic Systems Developer | Product Builder
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PROFESSIONAL SUMMARY

Software engineer (B.E. Information Science, 2027) focused on backend architecture and AI developer tooling. Backend and system architect for VMR Mangoes: designed its Node.js, Express and PostgreSQL 16 backend, with transactional inventory, idempotent Razorpay payments, role-based access and 900+ automated tests. Builds MCP-based code impact analysis for AI coding agents and Transformer-based demand forecasting. Winner, DevDays Bengaluru Build OpenAI Hackathon.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL
Backend: Node.js, Express.js, FastAPI, REST API design, Authentication & authorisation (JWT, RBAC, TOTP), Webhooks & idempotency
Databases: PostgreSQL, MySQL, SQLite, Schema design & migrations, Transactions & locking
Frontend: React, Next.js, HTML, CSS
AI / ML: PyTorch, Transformers (time series), Scikit-learn, Pandas, NumPy, Feature engineering, Model evaluation
AI agents: Model Context Protocol (MCP), AI agent integrations, Dependency graph analysis
Infrastructure & tooling: Git, GitHub Actions (CI/CD), Vercel, Neon, Docker, Playwright, Linux, Postman

PROFESSIONAL EXPERIENCE

Backend Architect & System Architect Present
VMR Mangoes, Bengaluru, India

- Architected the complete VMR Mangoes backend as a modular monolith on Node.js, Express and PostgreSQL 16, serving the storefront API, staff operations API and Razorpay payments from one service and one database.
- Designed a transactional stock engine in which time-limited holds, payment confirmation and FEFO batch allocation commit in a single PostgreSQL transaction, using row-level locks (FOR UPDATE, SKIP LOCKED) to prevent overselling.
- Built idempotent payment handling: HMAC-SHA256 webhook verification with timing-safe comparison, replay-safe payment outcomes, idempotency keys on refunds and automatic refunds of duplicate or wrong-amount payments.
- Implemented four-role access control (manager, operator, packer, QC) with TOTP two-step sign-in for staff and an activity log recording every staff change.
- Covered the backend with 900+ automated test cases and the React admin panel with Playwright end-to-end, axe accessibility and API-contract tests.

Software Engineering Intern Present
VMR Cloud Solutions, Bengaluru, India

Digital Content Producer & Brand Communications Associate Present
Yashoda Elevators, Bengaluru, India

- Conceptualise promotional campaigns and brand stories for the company's digital channels.

PROJECTS

AMU: Anti Messup Agent
Model Context Protocol (MCP), Dependency graphs, Static code analysis

- Built an AI-agent-oriented code impact analysis system that assesses the downstream effects of a proposed repository change before an agent makes it.
- Designed a repository dependency graph tracing direct and transitive relationships between functions and modules, exposed to AI coding agents as MCP tools with source-backed dependency paths.
- Surfaced related tests and documentation for targeted validation, with coverage and uncertainty information so agents can tell evidence from gaps.

Aurora
Python, PyTorch, FastAPI, Next.js 14, MySQL

- Architected an end-to-end multi-SKU demand forecasting pipeline around a Transformer time-series model in PyTorch (30-day input, 5 features, 7-day horizon).
- Turned forecasts into automated safety stock and reorder point calculations served through a FastAPI inference API and a Next.js 14 interface.

EDUCATION

Bachelor of Engineering, Information Science and Engineering Expected 2027
CMR Institute of Technology, Bengaluru

AWARDS & ACHIEVEMENTS

- Winner, DevDays Bengaluru Build OpenAI Hackathon. Top 10 of approximately 1,000 participating teams.
- Shortlisted, ISRO Robotics Challenge (IRoC-U). Autonomous UAV navigation and obstacle detection.
- First Place, Flex It Fitness Competition. Cash prize for outstanding performance.